Compulsory Assignment 2

Group 97 - Predictive Analytics (KAN-CDSCO1005U)

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Part 1

1.1 Time Series Plot, ACF, and Decomposition

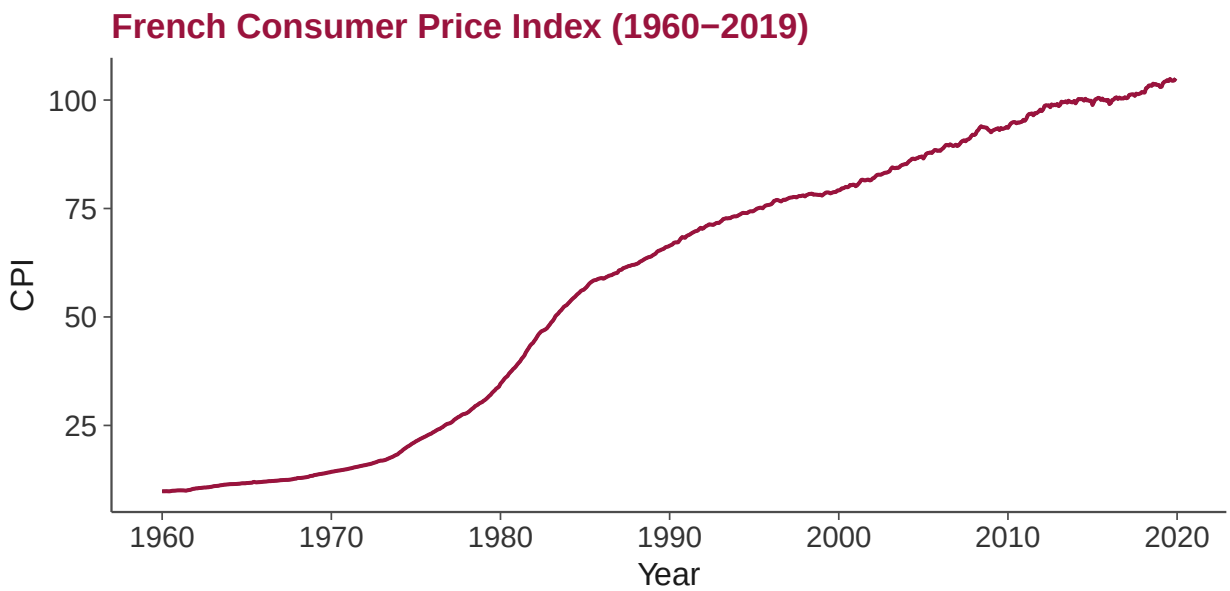

Figure 1: French CPI from January 1960 to December 2019.

The French CPI series displays a strong upward trend across the full sample period with no signs of mean reversion. The level rises from approximately 10 in 1960 to over 100 by 2019. Growth is steeper in the 1970s and early 1980s, likely driven by the oil crises and accompanying inflation, and flattens noticeably from the mid-1980s onward. A mild seasonal pattern is visible, particularly in the later decades. The variance of the series increases with its level, suggesting that a variance-stabilising transformation may be warranted.

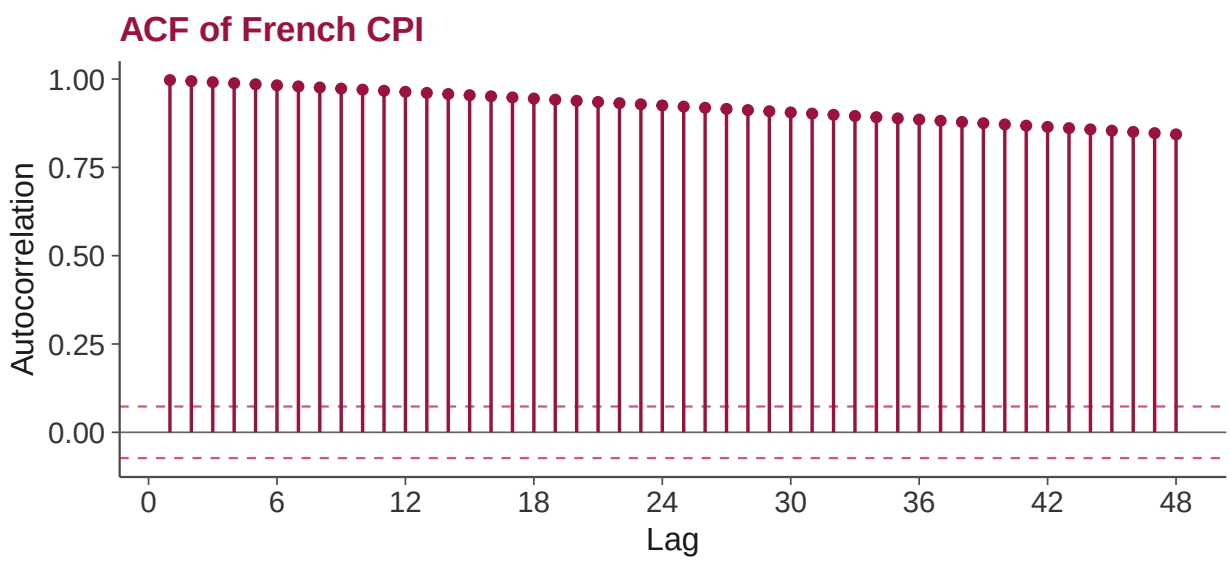

Figure 2: Sample autocorrelation function of the raw CPI series up to 48 lags.

The ACF decays extremely slowly, remaining significant well beyond 48 lags. This is characteristic of a non-stationary series with a strong trend component. The persistence confirms that the raw CPI level must be differenced or detrended before any ARIMA modelling.

We choose a classical multiplicative decomposition because the seasonal amplitude visibly grows with the level of the series, making a multiplicative specification more appropriate than an additive one. The classical method

is straightforward and gives a clear separation of trend, seasonal, and irregular components, which is useful as a first exploratory step.

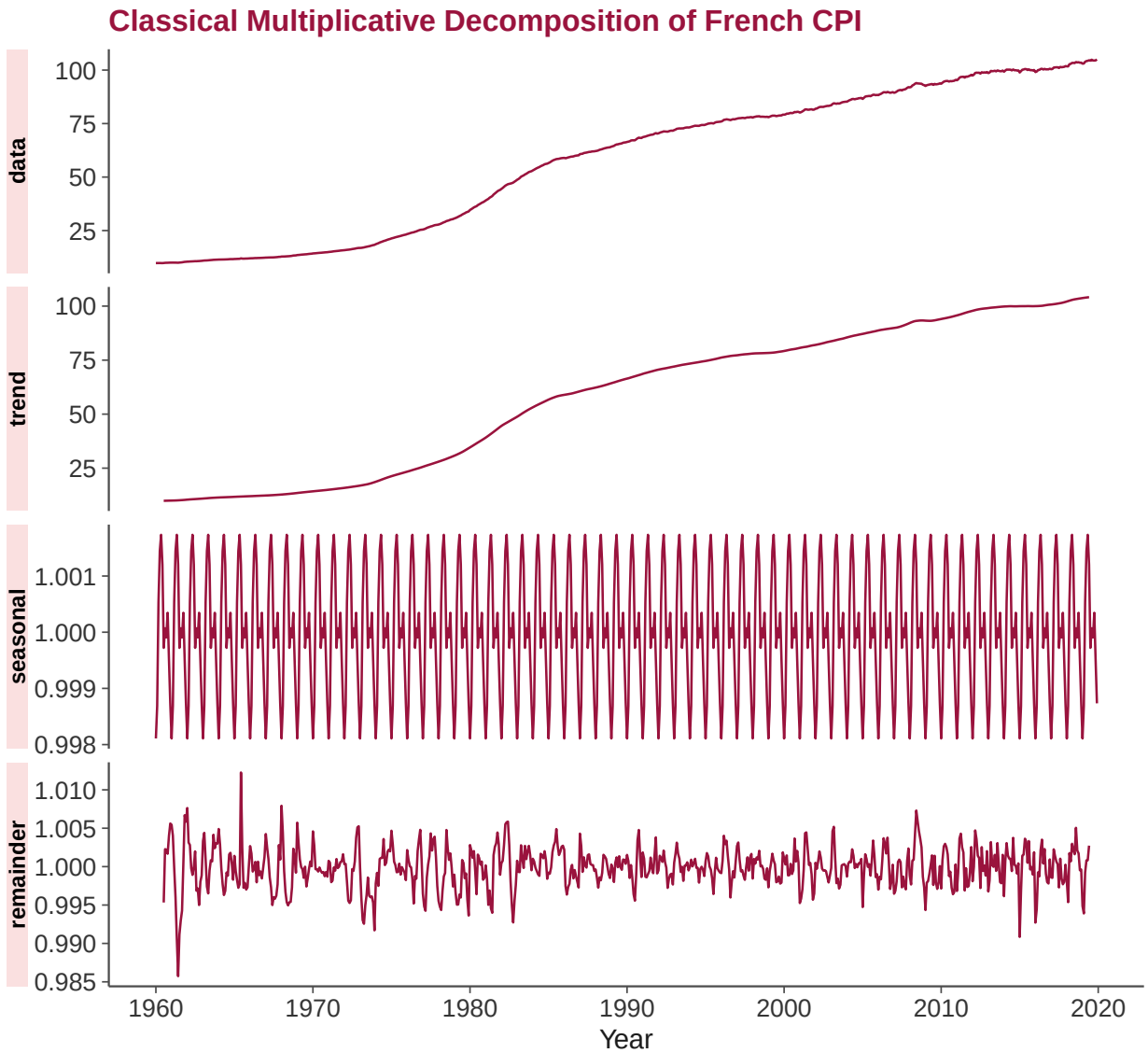

Figure 3: Classical multiplicative decomposition into data, trend, seasonal, and remainder components.

The decomposition reveals: (i) a smooth upward trend that captures the long-run inflationary movement, (ii) a seasonal component with a regular annual pattern, and (iii) an irregular (remainder) component that is largest during the high-inflation period of the 1970s.

Limitations of classical decomposition. The method uses symmetric moving averages, so the trend is unavailable for the first and last six months. It assumes that the seasonal pattern is perfectly constant over time, which is unlikely across a 60-year span where the structure of the economy changed considerably. It also offers no built-in mechanism for handling outliers or structural breaks, meaning that large shocks (e.g. the oil crises) feed directly into the remainder without adjustment.

1.2 Transformation

To address the increasing variance observed in the raw series, we consider two transformations: the natural logarithm and the Box-Cox transformation. The Box-Cox method selects the power parameter λ that best stabilises variance across the series. We use the Guerrero method implemented in `BoxCox.lambda()` to find the optimal λ.

The optimal Box-Cox λ is 1.3226. Since this is close to zero, it confirms that the natural logarithm ($\lambda = 0$) is a near-optimal variance-stabilising choice.

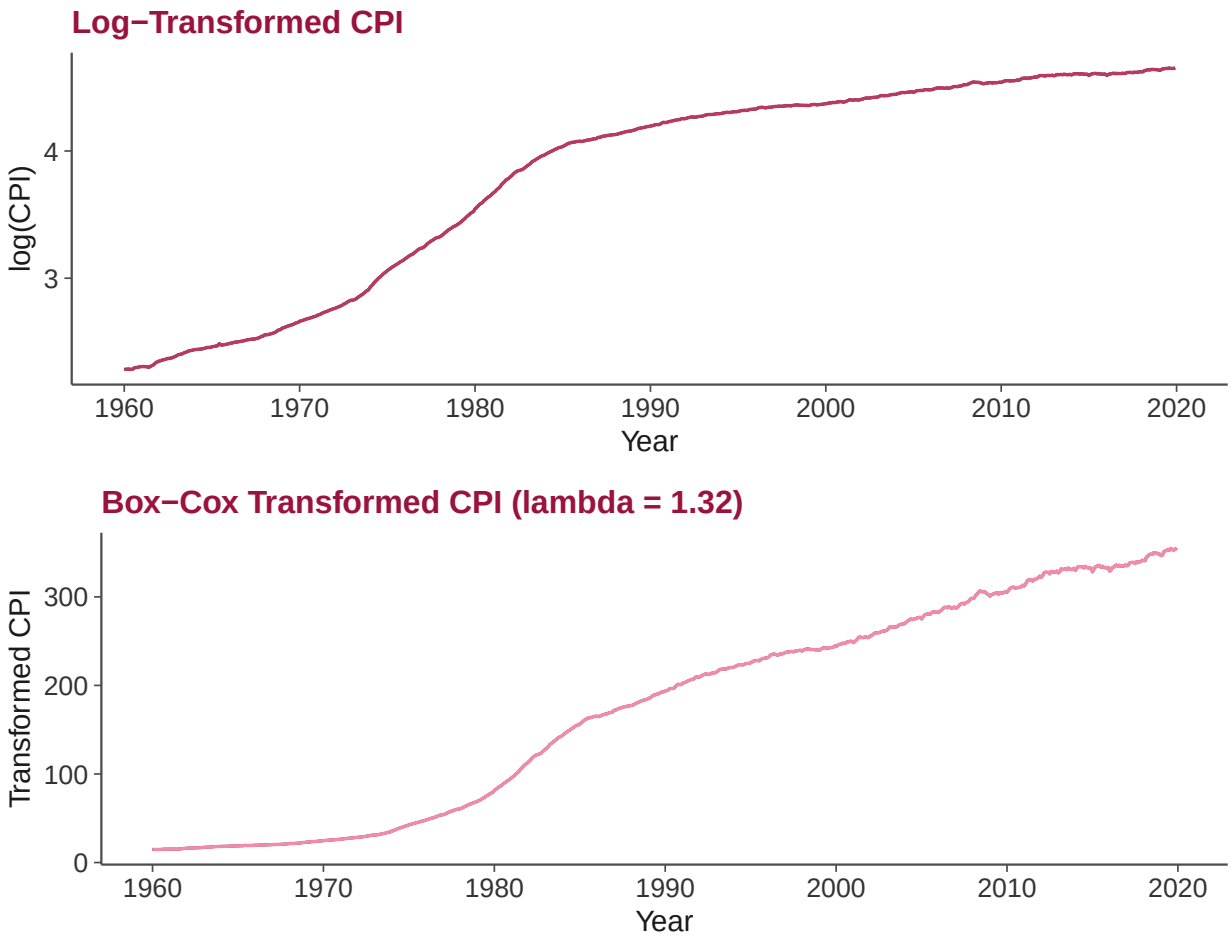

Figure 4: Comparison of log and Box-Cox transformations applied to the CPI series.

After the log transformation, the variance of the series is substantially more stable over time. We proceed with the log-transformed series for the remainder of the assignment because $\lambda \approx 0$ and the natural logarithm is simpler to interpret: first differences of log(CPI) approximate percentage changes, making economic interpretation straightforward.

With the log transformation applied, an additive decomposition is now appropriate because the seasonal fluctuations become approximately constant in magnitude.

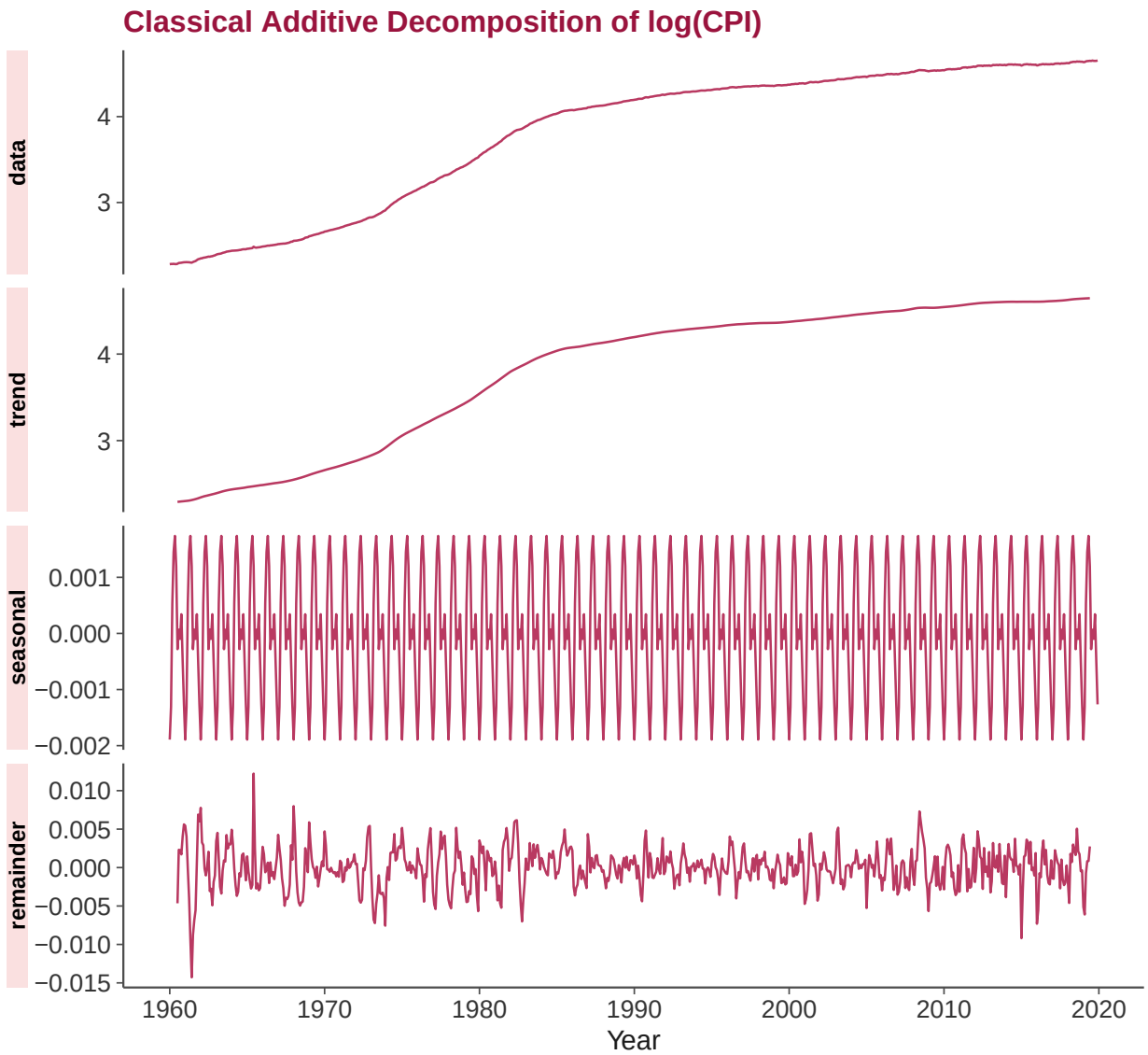

Figure 5: Additive decomposition of the log-transformed series.

The trend component remains strongly upward but is smoother in its curvature, and the remainder is more homoscedastic than in the original series. This confirms that the **log-transformed series is used for the remainder of the assignment**.

1.3 Unit Root Tests (KPSS and ADF)

The log-transformed CPI displays both a clear deterministic trend and a level shift, so the appropriate test specifications include a trend term. For the KPSS test, we use `type = "Trend"` to test stationarity around a deterministic trend. For the ADF test, we set `k = 12` to account for the monthly serial correlation structure, with the alternative hypothesis being trend stationarity.

Table 1: Unit root tests on log(CPI).

Test	Test Statistic	p-value
KPSS (Trend)	2.4629	0.01
ADF	-1.8044	0.661

Table 1: Unit root tests on log(CPI).

The KPSS test has the null hypothesis of (trend) stationarity. The p-value is below 0.01, so we reject the null: the series is not stationary even after accounting for a deterministic trend. The ADF test has the null hypothesis of a unit root. The large p-value means we cannot reject the null. Both tests consistently indicate that log(CPI) contains a unit root and requires differencing.

1.4 Making the Series Stationary and ARIMA Identification

Since the unit root tests confirm a stochastic trend and the decomposition reveals seasonality, we apply first-order regular differencing ($d = 1$) to remove the trend and first-order seasonal differencing ($D = 1$, at lag 12) to remove the seasonal pattern.

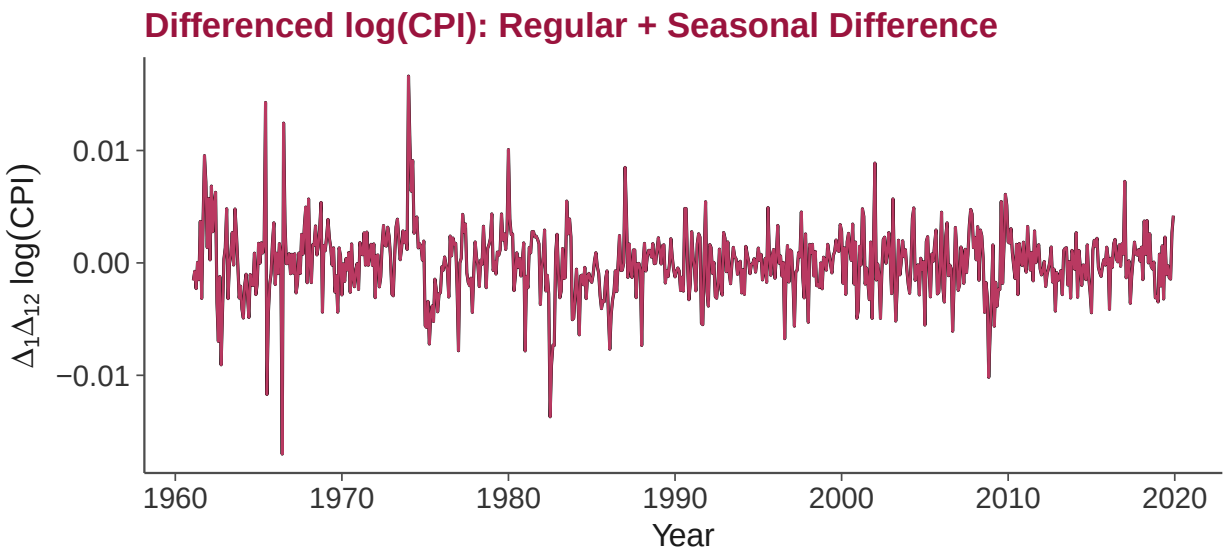

Figure 6: First regular and first seasonal difference of log(CPI).

After differencing, the KPSS test statistic is 0.1268 with a p-value of 0.1, so we do not reject the null of level stationarity. The series now fluctuates around zero with no visible trend or changing variance, confirming that it is stationary.

We now examine the ACF and PACF of the differenced series to identify candidate ARIMA orders for the non-seasonal and seasonal components.

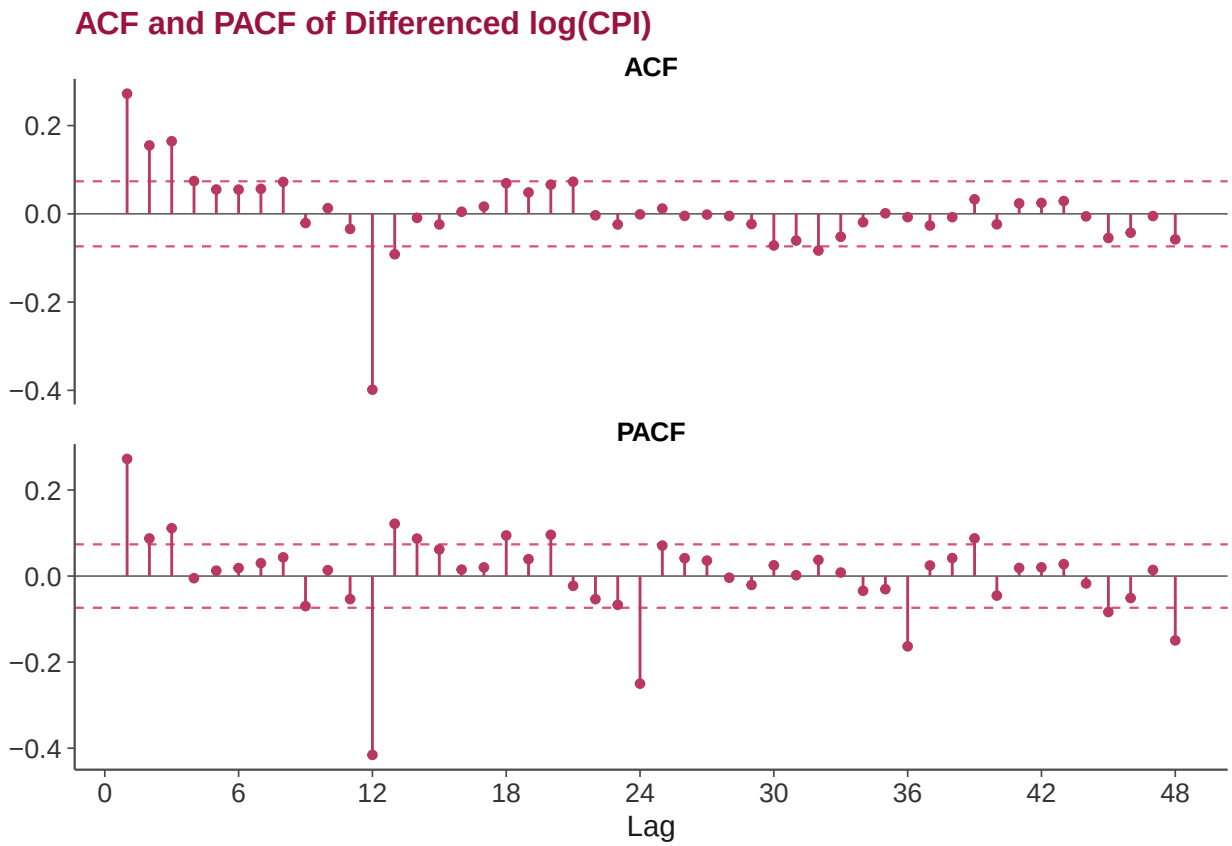

Figure 7: ACF (top) and PACF (bottom) of the doubly differenced log(CPI).

The ACF shows a significant spike at lag 1 that cuts off afterwards, suggesting a non-seasonal MA(1) component. At the seasonal lags (12, 24, . . .), the ACF shows a significant spike at lag 12 that decays, pointing to a seasonal MA(1) component. The PACF shows a significant spike at lag 1 with subsequent decay, which is consistent with the MA interpretation rather than an AR process.

Based on this analysis, we propose:

$$\text{ARIMA}(0,1,1)(0,1,1)_{12}$$

This is the classic "airline model" introduced by Box and Jenkins, which is well suited to economic series displaying a trend and multiplicative seasonality. In differenced form, the model can be written as:

$$(1-B)(1-B^{12})\log(Y_t) = (1+\theta_1 B)(1+\Theta_1 B^{12})\varepsilon_t$$

where B is the backshift operator, θ_1 is the non-seasonal MA coefficient, and Θ_1 is the seasonal MA coefficient.

Part 2

In this part, the log-transformed series is used throughout, as decided in 1.2.

2.1 Train/Test Split and Auto ARIMA

We split the sample so that the last 24 months (January 2018 to December 2019) serve as the test set. This provides two full years of out-of-sample observations while retaining 58 years of training data, which gives the model a substantial estimation window.

The training set contains 696 observations and the test set contains 24 observations.

We use `auto.arima()` with `ic = "aic"`, `stepwise = FALSE`, and `approximation = FALSE` to perform an exhaustive search over the ARIMA parameter space. This avoids the greedy stepwise shortcuts that can miss globally optimal specifications.

The AIC-selected model is **ARIMA(1,2,2)(2,0,0)[12]**. This differs from our manually identified ARIMA(0,1,1)(0,1,1)$_{12}$, suggesting that the automatic procedure detects additional short-run dynamics (an AR(1) term and a second MA term) as well as seasonal AR terms that improve the in-sample fit as measured by AIC.

To assess whether the model adequately captures the dynamics of the series, we examine the residuals. A well-specified model should produce residuals that resemble white noise: no significant autocorrelation, approximately normal distribution, and constant variance over time.

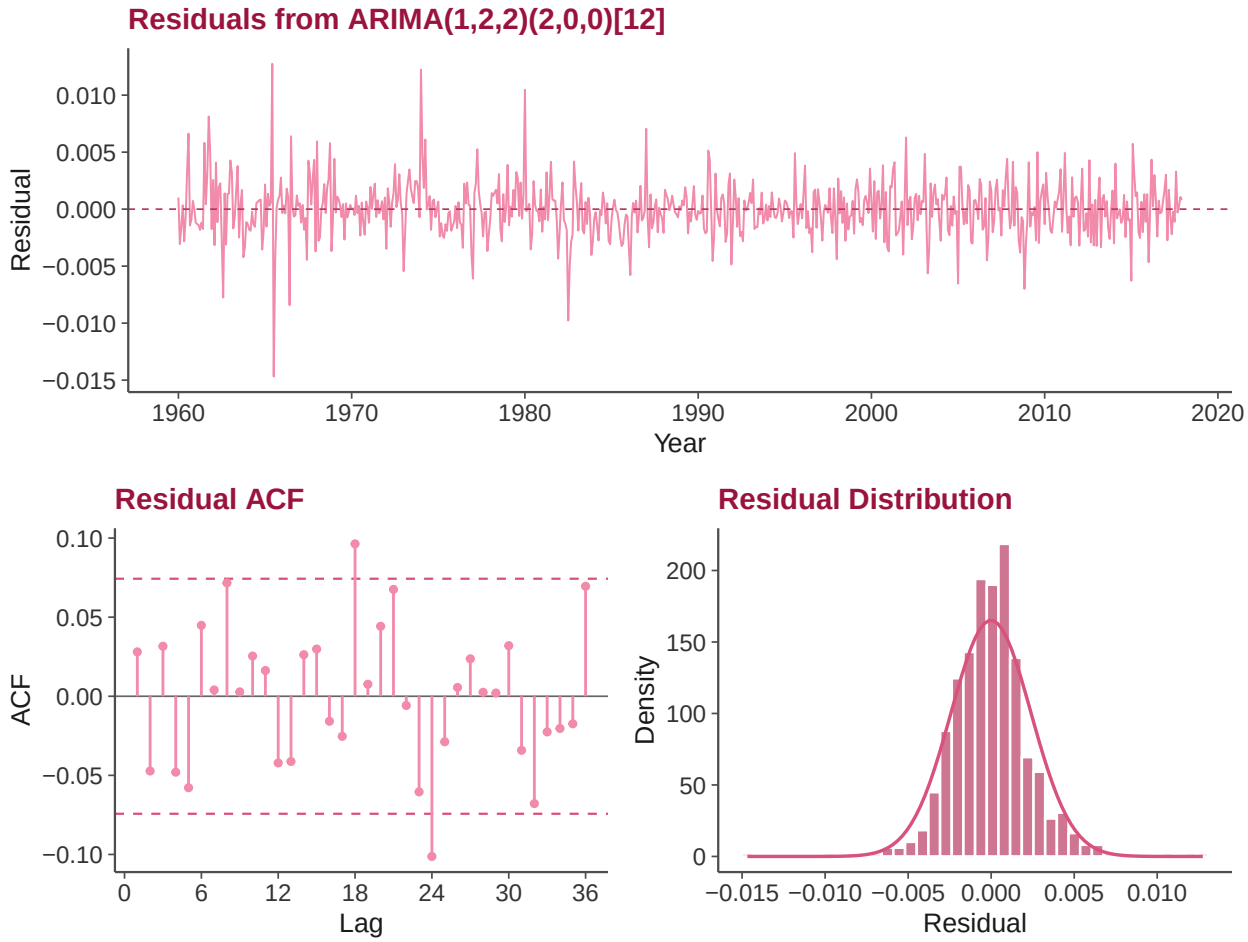

Figure 8: Residual diagnostics for the auto-selected ARIMA model. Top: time series of residuals. Bottom-left: residual ACF. Bottom-right: histogram with normal overlay.

We formally test for remaining autocorrelation with the Ljung-Box test at 24 lags, adjusting the degrees of freedom for the number of estimated parameters.

Table 2: Ljung-Box test on residuals of the auto-selected ARIMA model.

Test	Lags	X-squared	df	p-value
Ljung-Box	24	38.177	19	0.00563

Table 2: Ljung-Box test on residuals of the auto-selected ARIMA model.

The Ljung-Box p-value is below 0.05, indicating some residual autocorrelation at the 5% level. The residual time

series shows that the largest deviations concentrate in the 1960s and 1970s, periods of high inflation and structural change. The histogram is approximately bell-shaped but slightly heavy-tailed. The model could potentially be improved by adding further seasonal terms or by considering intervention variables for known structural breaks, but given the overall small magnitude of residuals the fit is reasonable.

2.2 Forecasting and Model Comparison

We now forecast over the 24-month test window using both models: the AIC-selected model from 2.1 and the manually identified ARIMA(0,1,1)(0,1,1)$_{12}$ from 1.4. The manual model is estimated on the same training set using `Arima()` with the specified orders.

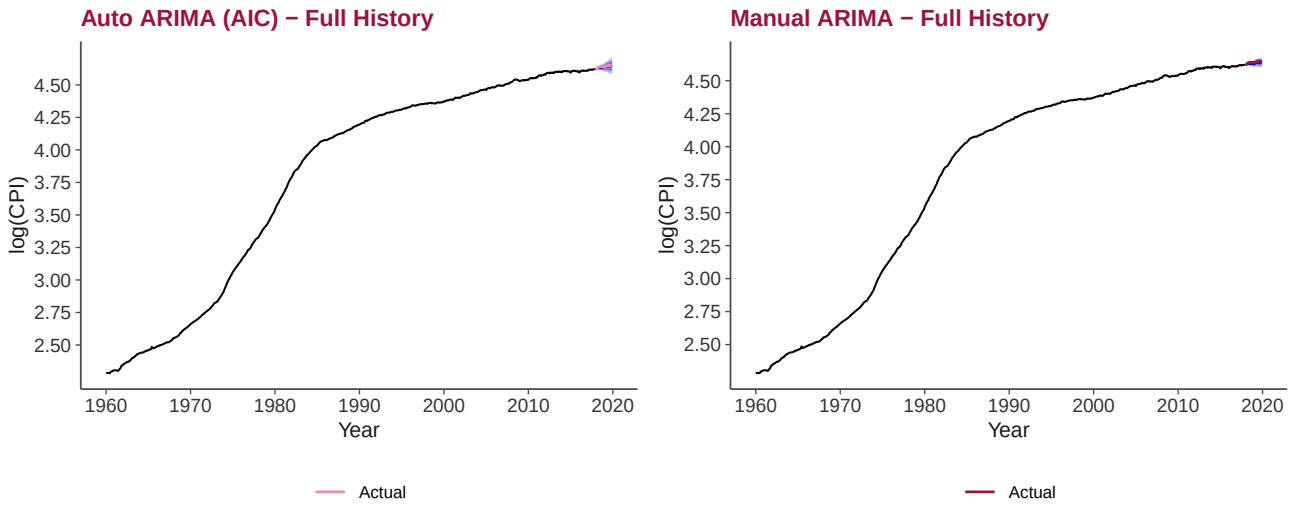

Figure 9: 24-month out-of-sample forecasts shown over the full sample history.

The full-history view confirms that both models track the training data well, but the forecast region is difficult to inspect at this scale. The zoomed view below focuses on 2015 onward to allow a clearer comparison of the prediction intervals and the actual test data.

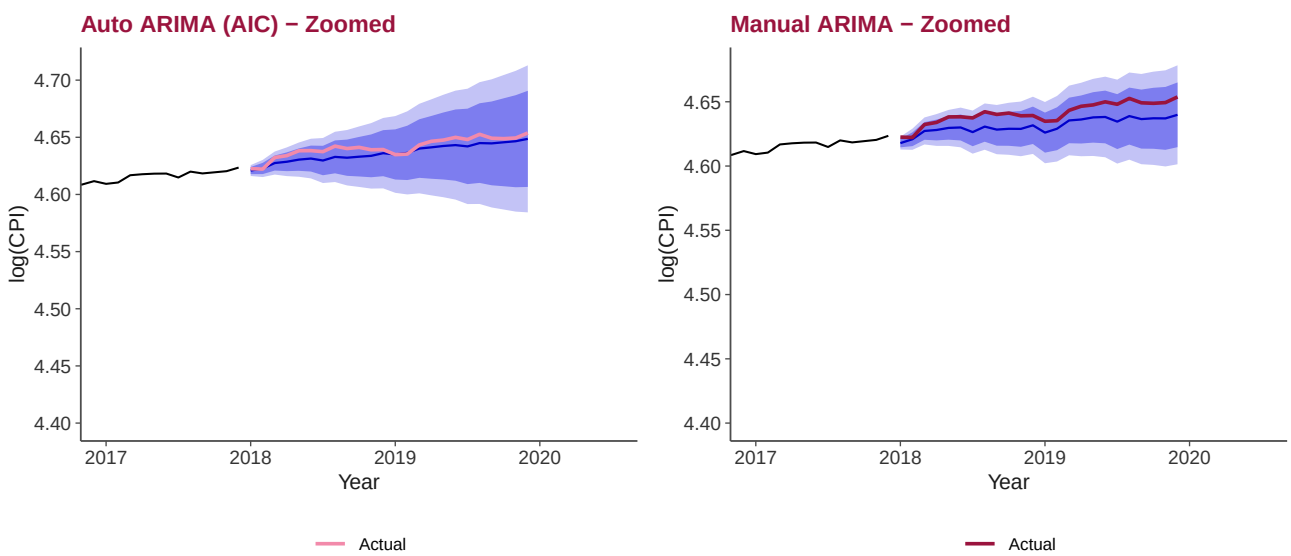

Figure 10: Zoomed forecast comparison (2015-2020). Shaded bands show 80 and 95 percent prediction intervals; the coloured line is the actual test data.

Table 3: Forecast accuracy: Auto ARIMA vs. Manual ARIMA(0,1,1)(0,1,1)[12].

Model	Set	RMSE	MAE	MAPE	MASE
Auto ARIMA (AIC)	Training set	0.0024	0.0017	0.0499	0.0426
Auto ARIMA (AIC)	Test set	0.0056	0.0050	0.1075	0.1220
ARIMA(0,1,1)(0,1,1)[12]	Training set	0.0025	0.0018	0.0522	0.0441
ARIMA(0,1,1)(0,1,1)[12]	Test set	0.0101	0.0096	0.2071	0.2350

Table 3: Forecast accuracy — Auto ARIMA vs. Manual ARIMA(0,1,1)(0,1,1)[12].

The accuracy table reports both in-sample (training set) and out-of-sample (test set) metrics. The Auto ARIMA model achieves a lower test-set RMSE and MAE, indicating that the additional parameters identified by the AIC search translate into better forecasting performance over the 24-month horizon. Both models have a MASE well below 1, meaning they comfortably outperform a naïve seasonal benchmark. However, the simpler airline model has the advantage of greater parsimony and interpretability, so the trade-off between marginal forecast improvement and model complexity should be considered depending on the intended use.